CH-5 LIFE PROCESSES
QUESTIONS BASED ON NUTRITION
BRAHMASTRA

1	**How do we tell the difference between what is alive and what is not alive?** Living things show features like movement, growth, reproduction, response to stimuli, while non-living things do not.	
2	**What criteria do we use to decide whether something is alive?** If something shows the following characteristics then it will be living thing otherwise it will be no living thing. • They can move on its own. • They need food to get energy and nutrients. • They respires. • They show growth and development. • They can breathe. • They remove their metabolic waste.	
3	**Is the invisible molecular movement necessary for life?** Yes, molecular movement are necessary for life because molecular movement powers life—fuelling energy, transporting nutrients, and maintaining balance. Without it, life would stop!	
4	**Why are molecular movements needed for life?** Molecular movements are essential for life because they enable nutrient transport, energy production, growth, communication between cells, and responses to the environment. Without these movements, vital biological processes like metabolism, repair, and adaptation wouldn't function, making life impossible.	
5	**What are the maintenance processes in living organisms?** **OR** **What processes would you consider essential for maintaining life ?** **N** Nutrition **C** Control and Coordination **E** Excretion **R** Respiration **T** Transport **G** Growth **M** Movement **are some processes that are essential for maintain life.**	
6	**What do you mean by life processes?** The basic essential activities performed by an organism for the survival are called life processes. Important life process include nutrition control and coordination excretion resolution transportation growth and movement etc.	
7	**What are the necessary conditions for autotrophic nutrition? And what are its by-products?** **Conditions:** Sunlight, chlorophyll, carbon dioxide, and water. **By-product:** Oxygen.	

8 **Where do plants get each of the raw materials required for photosynthesis?**

CO_2	From the atmosphere through stomata
Water	From the soil via roots
Sunlight	From the sun
Chlorophyll	Present in leaf cells

9 **Write any three points of differences between auto trophic and heterotrophic nutrition.**

AUTOTROPHIC NUTRITION

In autotrophic nutrition organisms prepared its own food.
They are not depended on any other organism for their food.
All green plants and certain bacteria are autotrophs.

HETEROTROPHIC NUTRITION

In heterotrophic nutrition organisms are not able to prepare their own food.
They are depended on other organism for their food.
All organism which are not included in the category of green plants are heterotrophs.

10 **Is nutrition a necessity for an organism? Discuss.**

Nutrition is necessary for an organism. It is required for the following purposes

- it provides energy for the metabolic process in the body
- it is necessary for the growth of new cells and repairing worn out cells.

11 **What are outside raw materials used for by an organism?**

All organism takes in oxygen, water and food as raw material from the outside.
As plants make their own food by the process of photosynthesis, they also need CO_2 and sunlight as raw material.

12 **What would happen if green plants disappear from earth?**

If green plants disappear, life on Earth will collapse. There will be no oxygen for breathing, no food for animals, and the entire food chain will break down. Carbon dioxide levels will rise, causing extreme climate changes. Eventually, all living beings will die due to starvation and lack of oxygen.

13 **What are the adaptations of leaf for photosynthesis?**

Leaves have several adaptations to maximize photosynthesis:

Large Surface Area	Helps absorb more sunlight.
Thin Structure	Allows quick diffusion of gases.
Chlorophyll Presence	Captures sunlight for energy
Stomata	Enable gas exchange (CO_2 in, O_2 out).
Veins	Transport water and nutrients.
Waxy Cuticle	Reduces water loss while allowing light in.

These features help plants efficiently to produce food and oxygen.

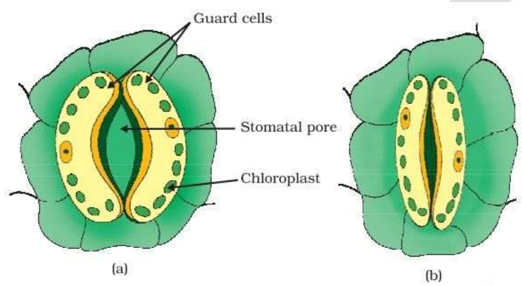
14 **How do we come to know that plants are alive?**

We see them green and they show growth and development.

15 **Why are the leaves of some plants not green in colour.**

If a plant appear of another colour such as red , purple it is not necessary because the plant does not contain chlorophyll it might be possible that other pigments may cover up the green pigment giving the plant a different colour other than green.

16	**How do non green plant do photosynthesis?** Photosynthesis also occur in plants which have non green leaves, chlorophyll is present in less quantity and other pigment mask the green colour of chlorophyll so they do not have green colour but carry out photosynthesis.	
17	**How do aquatic plants do photosynthesis** Aquatic plants get water and carbon dioxide from their aquatic environment and like the land plants light energy from the sun even though the plant is under water it is still get the sunlight energy from the sun because sunlight can pass through the water.	
18	**Why viruses are known as the connecting link between living and non-living things?** Virus are non-living and remains as a particle when not in contact with the host as soon as the virus enters the host body system it replicates like a living organism that's why they are considered connecting link between living and non-living things.	
15	**Write a balanced chemical equation for photosynthesis.** $6CO_2 + 12H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2 + 6H_2O$	
16	**List three events that occur during the process of photosynthesis.** 1. Absorption of light energy by chlorophyll. 2. Conversion of light energy into chemical energy and splitting of water molecules into hydrogen and oxygen. **3.** Reduction of carbon dioxide to carbohydrates (glucose) using hydrogen.	
17	**Explain the process of nutrition in amoeba.** Amoeba shows **holozoic nutrition**. The process involves the following steps: 1. **Ingestion:** Amoeba surrounds the food particle using pseudopodia and engulfs it. 2. **Digestion:** The food is enclosed in a food vacuole where digestive enzymes break it down. 3. **Absorption:** The digested food is absorbed into the cytoplasm. 4. **Assimilation:** The absorbed food is used for growth, repair, and energy. 5. **Egestion:** The undigested food is expelled through the cell membrane.	
18	**Explain the role of stomata in photosynthesis.** Stomata are tiny pores present on the surface of leaves. a) They help in gaseous exchange. b) They help in transpiration.	
19	**Leaves of a healthy potted plant were coated with Vaseline to block the stomata. Will this plant remain healthy for long? State three reasons for your answer.** No, the plant will not remain healthy for long. Stomata are needed for exchange of gases, photosynthesis, and transpiration. Blocking them with Vaseline prevents these processes, leading to the death of the plant.	
20	**Name the green dot like structures in some cells observed by a student when a leaf peel was viewed under a microscope. What is this green colour due to?** The green dot-like structures are chloroplasts. The green colour is due to the presence of chlorophyll pigment.	

21	**In the experiment "light is essential for photosynthesis ",why does the uncovered parts or the leaf turn blue black after putting Iodine solution ?** The uncovered part of the leaf receives sunlight and performs photosynthesis, producing starch. Iodine solution turns blue-black in the presence of starch.	
22	**What is common for cuscutta., ticks and leeches ?** Cuscuta, ticks, and leeches are parasites. They obtain food from a living host.	
23	**Name the tissue which transports soluble products of photosynthesis in a plant .** Phloem transports soluble products of photosynthesis (food) in plants.	
24	**Name the tissue which transports water and mineral in a plant .** Xylem	
25	**Draw a neat and well labelled diagram of open stomata and closed stomata.**  *(a) Open and (b) closed stomatal pore*	
26	**How is carbon dioxide obtained by (a) Aquatic plants (b) Terrestrial plants ?** (a) Aquatic plants obtain carbon dioxide dissolved in water. (b) Terrestrial plants obtain carbon dioxide from the atmosphere through stomata.	
27	**Define transpiration.** Transpiration is the loss of water in the form of water vapour from the aerial parts of a plant.	
28	**Do insectivorous plant perform photosynthesis? If yes, then explain why?** Yes, insectivorous plants perform photosynthesis because they contain chlorophyll. They trap insects only to obtain nitrogen, which is deficient in the soil where they grow.	
29	**How do desert plants perform photosynthesis.** Desert plants take up carbon dioxide at night and prepare an intermediate which is acted upon by the energy absorbed by the chlorophyll during the day.	
30	**Why do desert plants take up CO_2 at night?** Each time a plant opens its stomata, some water is lost. If this happens frequently during day time, high temperature will cause the water to evaporate quickly. To prevent this, the desert plants do not open their pores for carbon dioxide until the sun goes down.	
31	**How do guard cells regulate the opening and closing of stomatal pores?** Guard cells absorb water and swell up, causing stomata to open. When they lose water, they shrink and the stomata close.	
32	**In what form the nitrogen is taken up by the plants?** Plants take up nitrogen in the form of nitrates and nitrites dissolved in water.	

33	**Name the bacteria which converts atmospheric nitrogen into water soluble compounds like nitrates or nitrites.** Rhizobium bacteria	
34	**Is insectivorous plants heterotrophs or autotrophs?** Insectivorous plants are autotrophs because they prepare their own food by photosynthesis.	
35	**Although pepsin and trypsin are both protein digesting enzymes, they differ in action. Justify this statement by giving one difference between them.** Pepsin and trypsin both are protein digesting enzymes, but they differ in action. Pepsin acts only in acidic medium while trypsin acts in alkaline medium. Pepsin is secreted by the gastric glands and trypsin is secreted by the pancreatic juice.	
36	**Give the name of the enzyme present in the saliva and state the function of the enzyme. What would happen to the digestion process if this gland stopped producing this enzyme?** The enzyme present in the fluid in our mouth is called **salivary amylase**. Its function is to digest starch into sugar. If the salivary glands stopped producing salivary amylase, the initial digestion of starch in the mouth would be affected, making it more strenuous on the stomach and small intestine to digest these carbohydrates.	
37	**Give two reasons why bile juice is considered to be important during digestion in the small intestine.** Two reasons due to which secretion of liver are as follows: (i) The bile salts which are present in bile juice break down large globules of fat into smaller globules due to the deficiency of enzyme action. (ii) It makes the acidic food alkaline, so that pancreatic enzymes can act on it.	
38	**Name the site of complete digestion of the food in human body and name the end products formed of:** **(i) carbohydrates (ii) fats (iii) proteins** The site of complete digestion of carbohydrates, fats and proteins in the human body is the small intestine. (i) Glucose is the end product of carbohydrate. (ii) Fatty acids and glycerol are the end products of fats. (iii) Amino acid is the end product of proteins.	
39	**List two different functions performed by pancreas in our body.** The two different functions performed by pancreas in our body are as follows: (i) It secretes pancreatic juice, which contains enzymes like trypsin and lipase, which take part in the process of digestion. (ii) Beta cells of pancreas secrete insulin, which helps in controlling blood glucose levels.	
40	**Write in sequence the steps for experimental verification of the fact that "Sunlight is essential for photosynthesis".** Steps for experimental verification of the fact that "Sunlight is essential for photosynthesis". (i) Selection of a healthy plant (iv) Pluck and boil the covered leaf (ii) Cover a part of leaf to block sunlight (v) Treat with alcohol to remove chlorophyll (iii) Place the plant in sunlight (vi) Rinse and perform the iodine test **Result:** The uncovered parts of the leaf will turn blue-black, indicating the presence of starch and thus photosynthesis. The covered parts will remain unchanged, demonstrating that sunlight is necessary for photosynthesis.	

41	**What is the internal energy reserve in plants and animals?** The internal energy reserve in plants is **starch**, The internal energy reserve in animals, it is **glycogen**.	
42	**Why is diffusion insufficient to meet the oxygen requirements of multicellular organisms like humans?** In multicellular organisms all the cells may not be in direct contact with the surrounding environment. Thus, diffusion will not meet the requirements of all the cells.	
43	**In each of the following situations, what happens to the rate of photosynthesis?** **(i) Cloudy days** **Ans.** On cloudy days the rate of photosynthesis is decreased/reduced due to less availability of sunlight. **(ii) No rainfall in the area** **Ans.** In case of no rainfall in the area, rate of photosynthesis decreases. **(iii) Good manuring in the area** **Ans.** In case of good manuring, rate of photosynthesis increases due to increase in soil fertility. **(iv) Stomata gets blocked due to dust** **Ans.** When stomata gets blocked due to dust, the rate of photosynthesis decreases.	

PREPARED BY

-PRANAV TAYAL

These notes are specially prepared to make your learning simple, clear and effective.

Success doesn't come from luck, it comes from **dedication**, **consistency** and the **courage** to keep learning**.**

May god bless each one of you